

This figure displays a scatter plot comparing Predicted $\text{PM}_{2.5}$ ($\mu\text{g m}^{-3}$) on the y-axis against Observed $\text{PM}_{2.5}$ ($\mu\text{g m}^{-3}$) on the x-axis. The plot includes a dashed 1:1 line and a solid green calibration line. Marginal histograms are shown for both axes. Performance metrics are provided in a box:

- $R^2 = 0.732$
- $\text{RMSE} = 5.58 \mu\text{g m}^{-3}$
- $\text{MAE} = 4.06 \mu\text{g m}^{-3}$
- $\text{Bias} = -0.03 \mu\text{g m}^{-3}$

Legend:

- 1:1 line
- Calibration line

Figure 10b is a box plot showing the residuals (predicted minus observed) for $\text{PM}_{2.5}$ concentration ranges. The y-axis is labeled 'Residual (predicted - observed, $\mu\text{g m}^{-3}$)' and ranges from -30 to 30. The x-axis is labeled 'Observed $\text{PM}_{2.5}$ range ($\mu\text{g m}^{-3}$)' and has three categories: ≤ 20 (n=571), 20-35 (n=110), and > 35 (n=39). The plot shows three box plots: blue for ≤ 20 , orange for 20-35, and red for > 35 . Each box plot includes a median line, a box representing the interquartile range, and whiskers extending to the minimum and maximum values. A purple diamond indicates the mean bias for each category. The mean bias is approximately 0.5 for ≤ 20 , -2.5 for 20-35, and -5.0 for > 35 . The legend indicates that the purple diamond represents the 'Mean bias'.